CAPM Regression Model

The final class of diagnostics we will cover relates to the properties of the disturbance term ϵ_{it}.

The following properties need to be satisfied:

$$\mathbb{E}[\epsilon_{it}] = 0$$

1 $$\mathbb{E}[\epsilon_{it}^2] = \sigma_{i\epsilon}^2$$ ⇒ var ⇒ homoskedastic.

2 $$\mathbb{E}[\epsilon_{it}\epsilon_{i,t-k}] = 0, \quad k = 1, 2, \ldots$$ no autocorrelation

relationship ⇒ t $t-k$

If the above are not satisfied, CAPM standard errors may not be reliable.

Diagnostics

The **first** restriction tells us that idiosyncratic risk has zero mean.

The **second** property tells us that idiosyncratic risk, namely $\sigma^2_{i\epsilon}$, is constant across time.

1. This is known as homoskedasticity.
2. This assumption may hold for low frequency data, such as quarterly or yearly data, but tends to be violated for higher frequency data: daily, weekly, or monthly returns.

The **third** requirement states that movements in idiosyncratic risk are uncorrelated through time. This is known as NO autocorrelation.

LM Tests

- A *Lagrange Multiplier* (LM) test is a regression based diagnostic test.

- Operates on the following principle: If

$$\text{1} \qquad y_t = \alpha + \beta x_t + \epsilon_t$$

is correct, the residuals

$$\hat{\epsilon}_t = y_t - \hat{\alpha} - \hat{\beta} x_t$$

should be well-behaved.

- i.e., if the model is correct, $\hat{\epsilon}_t$ should **NOT** contain information about y_t that is not already captured by $\hat{\alpha} + \hat{\beta} x_t$.

- When there is some structure in the residuals that the model has missed, we say that the model is misspecified.

LM Tests in the CAPM

- For CAPM (applied to excess returns of asset *i*) to be well-specified:

$$\mathbb{E}[\epsilon_{it}^2] = \sigma_{i\epsilon}^2 : \text{Homoskedasticity}$$

$E(\varepsilon_1 \; \varepsilon_1)$

$= E(\varepsilon_1^2) \Rightarrow$ var.

no relationship across obs. ←

$$\mathbb{E}[\epsilon_{it}\epsilon_{il}] = 0 \; (t \neq l) : \text{No Autocorrelation}$$

$E(\varepsilon_1 \, \varepsilon_2) = 0$ $\quad$ $E(\varepsilon_2 \, \varepsilon_3) = 0$

don't match

- We can test each of these in turn.

- The condition $\mathbb{E}[\epsilon_{it}] = 0$ is satisfied by definition of the CAPM.

Homoskedasticity

- Homoskedasticity means that the disturbance term has a constant variance.
- This implies that the idiosyncratic risk does not vary over the sample.
- Heteroskedasticity translates into non-constant variance.
- Many different forms.

Homoskedasticity

- To test homoskedasticity we consider the **null and alternative-hypothesis**:

$$\mathrm{H}_0 : \mathbb{V}[\epsilon_{it}] = \sigma^2_{i\epsilon} \textbf{ is constant } \text{[Homoskedasticity]} \Rightarrow$$

s.e.(β_j) is correct.

$$\mathrm{H}_1 : \mathbb{V}[\epsilon_{it}] \textbf{ is not constant } \text{[Heteroskedasticity]} \Rightarrow$$

R output s.e.(β) is invalid.

- If heteroskedasticity is present, OLS standard errors are not correct.

- $\Rightarrow$ any conclusions for β_i or α_i in CAPM may be incorrect.

- LM test: captures sensitivities of the disturbance term to movements in the explanatory variables.

Homoskedasticity

- LM tests are based on the idea of a hypothesized regression model that explains variability in the residuals. For example,

var $\leftarrow$

1. $\hat{\epsilon}_{it}^2 = \gamma_{i0} + \gamma_{i1}x_t + \gamma_{i2}x_t^2 + v_{it}$;
2. $\hat{\epsilon}_{it} = y_{it} - \hat{\alpha}_i - \hat{\beta}_i x_t$
3. $y_{it} = r_{it} - r_{ft}$
4. $x_t = r_{mt} - r_{ft}$

- The **null hypothesis of homoskedasticity** can then be tested using

$$H_{i0} : \gamma_{i1} = \gamma_{i2} = 0$$

$\hat{\varepsilon}_{it}^2 = \gamma_{i0} \Rightarrow$ a number

- Denote the sample size by T.
- The **test statistic** used is

$$W = T \cdot R_i^2.$$

- **W** is asymptotically χ_2^2.

d.f. = 2 $\Rightarrow$ # of "equal" sign ≠ 0

Homoskedasticity

Let's consider testing for homoskedasticity in the three factor CAPM with tech. industry returns:

Step 1 $$(r_{it} - r_{ft}) = \alpha_i + \beta_{i1}(r_{mt} - r_{ft}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + \epsilon_{it}.$$

The auxiliary regression model is

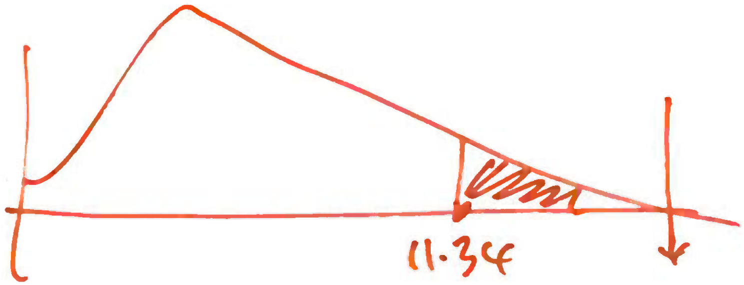
Step 2 $$\hat{\epsilon}_{it}^2 = \gamma_{i0} + \gamma_{i1}\underbrace{(r_{mt} - r_{ft})^2}_{X_1^2} + \beta_{i2}\underbrace{SMB_t^2}_{X_2^2} + \beta_{i3}\underbrace{HML_t^2}_{X_3^2} + v_{it}.$$

The test statistic is

$H_0: \gamma_{i1} = \beta_{i2} = \beta_{i3} = 0$

$$W = T \cdot R_i^2 = 1095 * 0.049 = 53.66, \quad > 11.34$$

and, under the null, this test has a χ_3^2 distribution. The critical value for the $\chi_3^2(.01) = 11.34$. Therefore, we **can reject the null** at the 1% critical value.



```
> res <- fit.tech$residuals

> white_aux <- lm(I(res^2) ~ I(MKT_RF^2)+I(SMB^2)+I(HML^2), data = data.xts)

> summary(white_aux)

Call:
lm(formula = I(res^2) ~ I(MKT_RF^2) + I(SMB^2) + I(HML^2), data = data.xts)

Residuals:
    Min      1Q  Median      3Q     Max 
-133.43   -7.26   -5.85   -1.04 1256.11 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  7.15217    1.35690   5.271 1.63e-07 ***
I(MKT_RF^2) -0.01896    0.01928  -0.984 0.325570    
I(SMB^2)     0.12611    0.02852   4.422 1.08e-05 ***
I(HML^2)     0.11687    0.03140   3.722 0.000207 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 42.35 on 1091 degrees of freedom
Multiple R-squared:  0.049,	Adjusted R-squared:  0.04639 
F-statistic: 18.74 on 3 and 1091 DF,  p-value: 7.425e-12


> white.test = summary(white_aux)$r.squared * nobs(white_aux)

> white.test
[1] 53.65732
```

Y

$R^2 \times T$

Results of Heteroskedasticity

- If we reject **homoskedasticity**, it does not tell us what form of heteroskedasiticty might be exhibited by the data.
- Only know that the OLS standard errors are incorrect.

Which Auxiliary Regression?

- We can consider different LM tests to detect **heteroskedasticity**.
- They do not tell us what form of heteroskedasiticty might be exhibited by the data.
- But what is the "right test"?
- No such thing. In general, just try to soak up as much variation as possible...Can do this via the option "White" in Eviews.

White's Testing Approach for Heteroskedasticity

- Exactly the same as earlier **LM test**.
- Just include all regressors, their squares and cross-products...
- Auxiliary regression:

$$\hat{\epsilon}_t^2 = \gamma_0 + \gamma_1 x_{1t} + \gamma_2 x_{2t} + \gamma_3 x_{1t}^2 + \gamma_4 x_{2t}^2 + \gamma_5 \underbrace{x_{1t} \cdot x_{2t}}_{\text{cross-product}} + v_t$$

- Usually works well.

Testing for Autocorrelation

- Autocorrelation means correlation through time.
- If autocorrelation is present, the standard errors given by OLS are not correct $\Rightarrow$ conclusions drawn about β_i or α_i may be incorrect.
- To test this, we can again consider an LM test with either auxiliary regression model

$p = 2, 3, 4$

no square

$$\hat{\epsilon}_{it} = \gamma_{i0} + \gamma_{i1}x_{1t} + \cdots + \gamma_{ik}x_{kt} + \rho_{i1}\hat{\epsilon}_{i,t-1} + \cdots + \rho_{ip}\hat{\epsilon}_{i,t-p} + v_{it},$$

t $\quad t-1 \quad t-p$

or

$$\hat{\epsilon}_{it} = \gamma_{i0} + \rho_{i1}\hat{\epsilon}_{i,t-1} + \cdots + \rho_{ip}\hat{\epsilon}_{i,t-p} + v_{it},$$

where v_{it} is the disturbance term.

no auto

$H_0: \rho_{i1} = \rho_{i2} = \cdots = \rho_{ip} = 0$

Testing for Autocorrelation

- We can then test **the null hypothesis of no autocorrelation** using

$$\begin{aligned} \mathrm{H}_0 &: \rho_{i1} = \rho_{i2} = \cdots = \rho_{ip} = 0 \text{ [No Autocorr.]} \\ \mathrm{H}_1 &: \text{ At least one differs from zero [Autocorr.]} \end{aligned}$$

- The **test statistic** for this null hypothesis follows the standard form given in this section:

$$\mathrm{AR}(p) = T \cdot R_i^2$$

- Under the null hypothesis, $\mathrm{AR}(p)$ is asymptotically χ_p^2.

Testing for Autocorrelation

Consider testing for autocorrelation in the three factor CAPM:

$$(r_{it} - r_{ft}) = \alpha_i + \beta_{i1}(r_{mt} - r_{ft}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + \epsilon_{it},$$

for tech. portfolio returns. We consider an auxiliary regression model

$p = 2$

$$\hat{\epsilon}_{it} = \gamma_{i0} + \gamma_{i1}(r_{mt} - r_{ft}) + \gamma_{i2}SMB_t + \gamma_{i3}HML_t + \rho_{i1}\hat{\epsilon}_{i,t-1} + \rho_{i2}\hat{\epsilon}_{i,t-2} + \nu_{it}.$$

The test statistic is

$H_0: \rho_{i1} = \rho_{i2} = 0$

$$\text{AR}(2) = \text{LM-test} = 9.362,$$

2 equal sign

and, under the null, this test statistic has a χ^2_2 distribution. The critical value for the $\chi^2_2(.01) = 9.21$. Therefore, we can **reject the null** at the 1% critical value.

d.f. = 2

Breusch-Godfrey or LM test

```
> lmtest::bgtest(fit.tech, order = 2, type = "Chisq")

	Breusch-Godfrey test for serial correlation of order up to 2

data:  fit.tech
LM test = 9.3622, df = 2, p-value = 0.009269
```

Week 6: Efficient Market Hypothesis (EMH) and Implications

CAPM: Diagnostics for the Disturbance Term (Continued)

What should we do now?

- We've seen that we can reject both the **null of homoskedasticity** and the **null of no autocorrelation** for the 3-factor CAPM model on technology portfolio returns.

- OLS standard errors for $\alpha_i, \beta_{i1}, \cdots, \beta_{i3}$ are not correct.

- What can we do in such cases?

- Point estimates are fine, need to adjust the standard errors.

- This can be done through the use of standard errors that are robust in the presence of heteroskedasticity and autocorrelation.

Newey & West Standard Errors

- Newey and West (1987) propose a covariance estimator that is consistent in the presence of both heteroskedasticity and autocorrelation (HAC) of unknown form.

  covariance.

- These standard errors can be calculated in Eviews easily.

- Let's see how to do this in the case of the 3-factor CAPM for the tech. portfolio.

HAC

```
ff3f.fit <- lm(data.xts$TECH_RF~data.xts$MKT_RF
                   +data.xts$SMB+data.xts$HML)
summary(ff3f.fit)

library(lmtest)
library(sandwich)

T <- nobs(ff3f.fit)
m <- floor(0.75 * T^(1/3))

# compute newey-west HAC uses library(sandwich)
NW_VCOV <- NeweyWest(ff3f.fit,
                     lag = m-1 , prewhite = F,
                     adjust = T)

# Std errors, t-test, p-value uses library(lmtest)
coeftest(ff3f.fit, vcov = NW_VCOV)
```

⇒ invalid s.e.(β_i)

⇒ N-W derive a formula for s.e.$(\hat{\beta}_i)$ assuming hetero and autocorelation

HAC

valid.

$\frac{1}{2}$

```
t test of coefficients:

                 Estimate Std. Error t value  Pr(>|t|)    
(Intercept)      0.200697   0.103077  1.9471   0.05178 .  
data.xts$MKT_RF  1.212908   0.031616 38.3637   < 2e-16 ***
data.xts$SMB     0.947409   0.077403 12.2399   < 2e-16 ***
data.xts$HML    -0.188331   0.072819 -2.5863   0.00983 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

same as before

HAC valid under hetero and auto.

new

valid.

Exchange Traded Funds

At the end of June each year, Fama and French sort eligible stocks into:

Size: Small (S) or Big (B), using the NYSE median market equity

Book-to-market: Low (L), Neutral (N), or High (H), using the 30th and 70th NYSE percentiles of BE/ME

Book → market

This creates 6 portfolios.

March 2026

	Book-to-market (BtM)			
	30% smallest Low	Neutral	70% largest High ← sort.	return (%)
either → Small or	S/L	S/N	S/H	1% 1.8 2.5
Big	B/L	B/N	B/H	0.6 1.1 1.7
	↓ growth		↓ value	

SMB

SMB = Small Minus Big

Take the average return on the three small-stock portfolios and subtract the average return on the three big-stock portfolios:

return

$$\mathrm{SMB} = \underbrace{\frac{1}{3}(S/L + S/N + S/H)}_{\text{first row}} - \underbrace{\frac{1}{3}(B/L + B/N + B/H)}_{1.13} = 0.6334\%$$

↓

March 2026

$\frac{1}{3}\underbrace{(1 + 1.8 + 2.5)}_{1.76} - 1.13$

So SMB asks:

On average, did small firms outperform big firms this period?

HML

HML = High Minus Low

Take the average return on the two high book-to-market portfolios and subtract the average return on the two low book-to-market portfolios:

March 2026

$$\text{HML} = \frac{1}{2}(S/H + B/H) - \frac{1}{2}(S/L + B/L) = 1.33\%$$

So HML asks:

Did value stocks (high BE/ME) outperform growth stocks (low BE/ME)?

Quant (factor) investing

Such factor models have also huge practical applications.

Many ETFs target these factors directly (Quant (factor) investing).

Moneysmart .gov.au

Exchange traded funds (ETFs)

Exchange traded funds (ETFs) can be a low-cost way to earn a return similar to an index or a commodity.

market index - SP500

small stocks index

Vanguard Investments Superannuation Adviser solutions

Do you not like money?

Start investing in future you with Vanguard's low-cost investments and low-fee super.

VUG Vanguard Growth ETF

Overview | Performance & fees | Price | Portfolio composition

Key facts

IOV ticker symbol	VUG.IV
CUSIP	922908736
Management style	Index
Asset class	Domestic Stock - General
Category	Large Growth
Inception date	01/26/2004

VV Vanguard Large-Cap ETF

Overview | Performance & fees | Price | Portfolio compositio

Key facts

IOV ticker symbol	VV.IV
CUSIP	922908637
Management style	Index
Asset class	Domestic Stock - General
Category	Large Blend
Inception date	01/27/2004

Neutral

For US-listed Vanguard ETFs:

- **Large growth:** VUG
- **Large blend/core:** VV
- **Large value:** VTV
- **Mid growth:** VOT
- **Mid blend/core:** VO
- **Mid value:** VOE
- **Small growth:** VBK
- **Small blend/core:** VB
- **Small value:** VBR

o.k.

Week 3 Workshop: This is the classic Fama table.

big gap

		growth				value
				BtM		
		Low	2	3	4	high
	small	0.30	0.81	0.85	1.03	1.17
	2	0.48	0.73	0.93	0.96	1.04
size	3	0.50	0.79	0.79	0.89	1.09
	4	0.60	0.57	0.72	0.87	0.87
	Big	0.46	0.52	0.49	0.57	0.65

%

Drawing on the Fama–French framework, let's consider the Vanguard style-box table - why not?

	Growth	Blend	Value
Small	VBK	VB	VBR
Mid	VOT	VO	VOE
Large	VUG	VV	VTV

ProShares ETF

Our ETFs Strategies Insights

SDS

UltraShort S&P500

Why SDS?

Seek to Profit from Downturns

SDS is the only -2x ETF designed to profit when the daily price of the S&P 500 declines.

Portfolio Hedging

SDS provides the opportunity to offset an expected market decline.

SDS is an ETF

Why SDS?

global crisis

Seek to Profit from Downturns

SDS is the only -2x ETF designed to profit when the daily price of the S&P 500 declines.

"minus" → win.

ProShares claims that SDS (ticker symbol: SDS) aims to provide approximately -2 times the daily return of the S&P 500 Index (`^GSPC`).

$-2 \times r_t$

Using the concepts learned in this unit, explain how you would evaluate whether ProShares UltraShort S&P500 (SDS) appropriately tracks its stated daily target.

To do this, download daily adjusted closing price data for SDS and daily closing price data for the S&P 500 Index (`^GSPC`) from Yahoo Finance, using a sample period that starts on 1 July 2006.

Efficient Market Hypothesis

idea: $\underbrace{\text{use past infor}}_{r_{t-1},\ r_{t-2},\ \ldots}$ to predict r_t

$r_t \downarrow$ today

$r_{t-1}, r_{t-2}, \ldots \downarrow$

Conditional on past info.

not on growth/ small.

Motivation

1. The goal of investing is to earn a profit.
2. The most meaningful measure of profit for investors is returns.
3. *Expectations* of returns are a key consideration. They affect asset demand and the price of securities.
4. A contentious topic in modern finance is whether or not returns are predictable:
   - Not the case if all available information about an asset is automatically accounted for in its price, and thereby its return.
   - If returns are predictable, it would be possible to obtain a positive return on some asset without any risk.
   - You could, in a sense, "beat" the market.

using past info

yesterday, r_{t-2}, r_{t-3}

Motivation

no relationship btw today and yesterday.

- The idea that the market can not be "beaten" is sometimes referred to as the **Efficient Markets Hypothesis** (EMH). ⇒ return not predictable.
- **Fama**'s 1965 and especially 1970 papers made EMH the standard framework in modern finance.

EMH

If true, the EMH implies that:

1. The current price of an asset reflects all information available in the market.
2. The current price provides no information regarding future asset movements.
3. Future returns are completely unpredictable, given information on past returns.
4. Traders can not systematically use newly arriving information to make a profit.
5. Conditional on all previous information, returns are completely random.

Forms of EMH

- What are the forms of EMH?

- The **weak form** test: (or random walk theory): prices reflect all the
feasible → easy to do information in past prices => future changes in stock prices should be unpredictable.

- The **semi-strong form**: prices reflect all publicly available information => if information is publicly available, then a positive announcement about a company will not, on average, raise the price of its stock because this information has already been *priced in*.

- The **strong form**: prices reflect all acquirable information.

Testing EMH

To understand whether EMH is satisfied, we must consider a broad econometric framework that allows us to test for satisfaction of the EMH.

return r_t

Let y_t be our variable of interest (this could be returns, but need not necessarily be returns).

$$F_{t-1} = r_{t-1}, r_{t-2}, \ldots\ldots$$

all info up until "t-1"

conditional on F_{t-1}

Define $\mathcal{F}_{t-1}$ to be all information available to the econometrician at time $t-1$.

This could consist of possible lagged values of y_t, such as $y_{t-1}, y_{t-2}, \dots, y_1$, or additional variables such as covariates.

To test the EMH, we will use time series models and methods.

idea : predict r_t using F_{t-1} → all info

t today

$t-1$ (up until)

$I_{t-1} \Rightarrow F_{t-1}$

"error"

A time series $\{\epsilon_t\}_{t \geq 1}$ is called a **white noise** process if

WN

$\gamma(1) = 0$

$\gamma(2) \neq 0$

$\downarrow$

ϵ_t is NOT WN

1. $\mathbb{E}[\epsilon_t] = 0$;

2. $\mathbb{V}[\epsilon_t] = \sigma_\epsilon^2$ for all t; $\Rightarrow$ a number

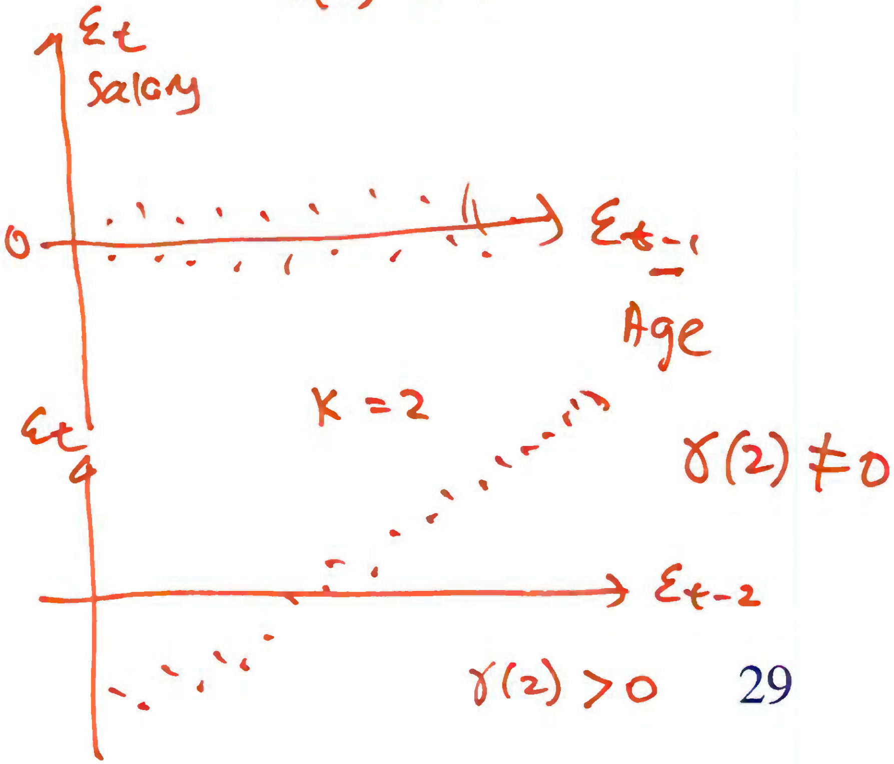
Focus $\Rightarrow$ 3. $\gamma_\epsilon(k) = \text{Corr}(\epsilon_t, \epsilon_{t-k}) = 0$, for all t and $\underline{k}$.

$k=1$

$\gamma(1) = 0$

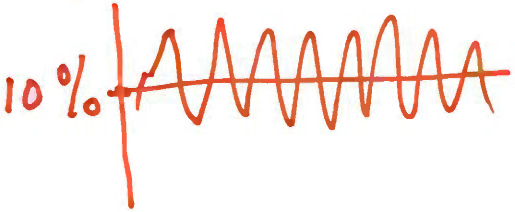
This is sometimes written as $\epsilon_t \sim \text{WN}(0, \sigma_\epsilon^2)$.

	ε_t	ε_{t-1} ($k=1$)	ε_{t-2} ($k=2$)
$t=1$	$\varepsilon_1 = 1.5$	N.A. ε_0	N.A $= \varepsilon_{-1}$
$t=2$	$\varepsilon_2 = 1$	$\varepsilon_1 = 1.5$	N.A ε_0
$t=3$	$\varepsilon_3 = 0.5$	$\varepsilon_2 = 1$	$\varepsilon_1 = 1.5$
	$\vdots$	$\vdots$	$\vdots$
	salary	~~Age~~ Age	X_2 educ



r_t A model for log returns

no X variable



intercept $\Rightarrow$ $\underbrace{\hat{\mu} = T^{-1} \sum_{t=1}^{T} r_t}_{\text{sample mean}}$ $= 10\%$

Suppose

to de-mean so that ϵ_t has zero mean $\Leftarrow$

$$r_t = \mu + \epsilon_t,$$

and $\epsilon_t \sim \mathrm{WN}(0, \sigma_\epsilon^2)$, then the EMH would be satisfied.

$E(\epsilon_t) = 0 \Rightarrow$ has zero mean.

Even if we knew ϵ_t, it would not help us predict, on average, the future values of r_{t+h}, $h \geq 1$.

WN $\Rightarrow$ $\left.\begin{array}{l} \gamma(1) = 0 \\ \gamma(2) = 0 \\ \vdots \\ \gamma(k) = 0 \quad \text{for any } k \end{array}\right\}$ No relationship whatsoever.

$\boxed{\text{EMH holds.}}$

If we assume the term ϵ_t is white noise, we can test the EMH hypothesis in two ways:

Method 1: Return predictability mean

If markets are efficient then returns, at any frequency, should not be predictable.

Method 2: To compare the variance of asset returns over different time horizons.

If markets are efficient, then the variance of asset returns over different time horizons should roughly increase with their horizon. If this is the case, the variance of n period returns should simply be n times the variance of the 1-period return. Slide 58

Measure return predictability through serial dependence. The most common measure of **sample serial dependence** is (sample) autocorrelation function (ACF).

The sample "autocorrelation function" is given by:

hat

$$\hat{\gamma}(k) = \frac{\sum_{t=k+1}^{T}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{T}(r_t - \bar{r})^2}. = \frac{\text{cov}(\text{"}t\text{"}, \text{"}t-k\text{"})}{\text{var}}$$

$k = 1$ $\quad \hat{\gamma}(1)$ = what is this number?

(1) r_t and r_{t-1} not related

(2) r_{t-1} is useless info. knowing r_{t-1} is not useful.

(3) $\hat{\gamma}(1) = 0$

(3) r_t is not predictable.

(4) Fama is correct

(5) EMH holds.

(6) r_t forgets. $\Rightarrow$ $\hat{\gamma}(1) = 0$

The population ACF is

$$\gamma(k) = \frac{\text{Cov}(r_t, r_{t-k})}{\mathbb{V}(r_t)}.$$

$\Rightarrow$ auto - correlation

$k=1,\ k=2,\ k=3 \ldots$

$\gamma(1)$
$\gamma(2)$
$\gamma(3)$
$\vdots$

$k=1$

$-1 \leq \gamma(k) \leq 1$

$\gamma(1) = 1 \Rightarrow$ remember everything.

$\gamma(1) = 0 \Rightarrow$ forget " $\Rightarrow$ EMH

$\gamma(1) = 0.5 \Rightarrow$ remember something

1:10

1:15 back.

Autocorrelation function (ACF)

hat

1. $\hat{\gamma}(k)$ measures the strength of association (correlation) between returns at time t and returns at time $t - k$.

2. Numerator of $\hat{\gamma}(k)$ is just the covariance of r_t and r_{t-k}.

3. Denominator is a normalization to ensure that $\hat{\gamma}(k)$ is between -1 and 1.

Return Predictability

1. If returns exhibit no predictability, then there should be no autocorrelation, as returns at any period t would not have any correlation with returns at period $t - k$.

2. If returns are not predictable, $\hat{\gamma}(k) \approx 0$ for any $k = 1, 2, \ldots$.

not zero.

3. If returns exhibit positive or negative autocorrelation, then this pattern can be exploited to predict future behavior, and hence the EMH would not be satisfied.

Examples

Example 1: Consider the hourly observations for the period 00:00 1 January 1986 to 11:00 15 July 1986 on the \$ / £ (ie. U.S. dollars per pound, how many US dollars are needed to buy one British pound) exchange rate.

exchange rate : Yes / No

Example 2: Consider the daily prices for the U.S. Apple stock from 14 March 2005 to 13 March 2014.

Yes / No

stock market

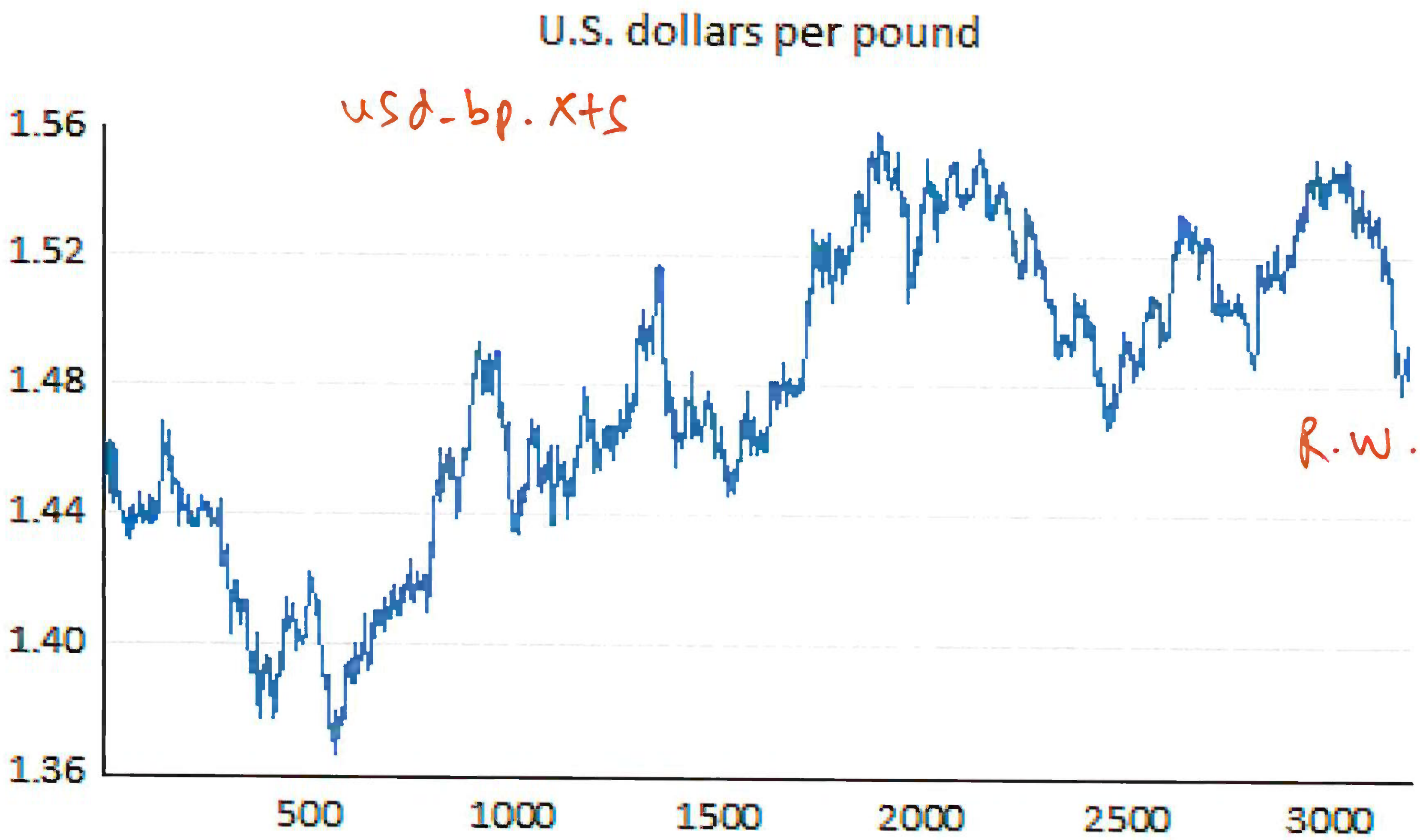
Q : EMH ⇒ hold or not ?

U.S. dollars per pound

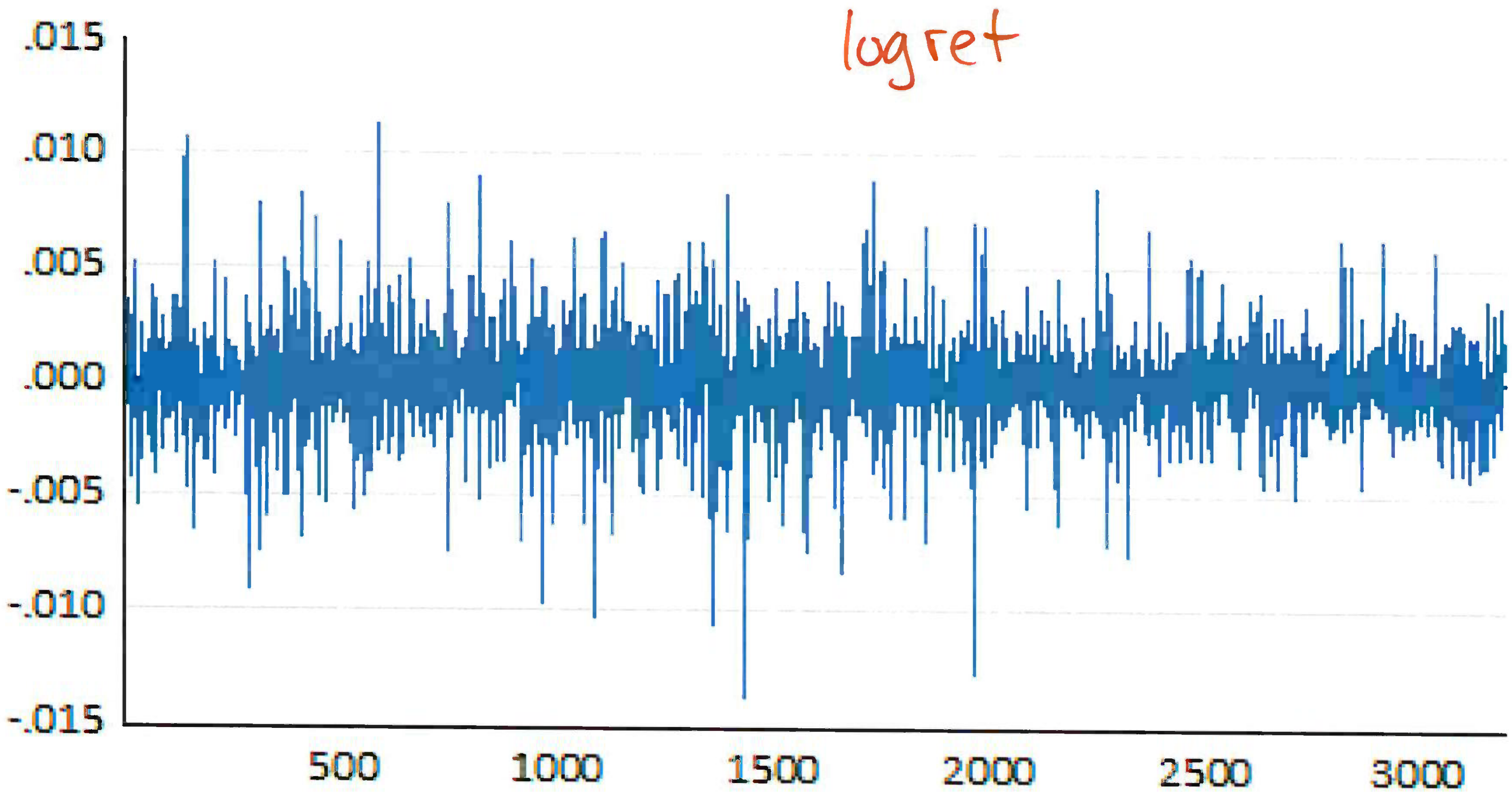
```
# plot U.S. dollars per pound
plot(usd_bp.xts)

# log return
logret <- na.omit(diff(log(usd_bp.xts)))

# plot log return
plot(logret, main = "log return")
```


U.S. dollars per pound
usd-bp.xts
1.56
1.52
1.48
1.44
1.40
1.36
500
1000
1500
2000
2500
3000
R.W.

log return


logret
.015
.010
.005
.000
-.005
-.010
-.015
500
1000
1500
2000
2500
3000

```
> source("Correlogram.R")

> length(logret)
[1] 3189

> correlogram(logret,max.lag = 10)
 Lag      AC       Q      Prob
   1 -0.0610 11.8834 5.664e-04
   2 -0.0013 11.8891 2.620e-03
   3  0.0202 13.1934 4.237e-03
   4 -0.0318 16.4287 2.495e-03
   5  0.0318 19.6576 1.449e-03
   6 -0.0584 30.5479 3.092e-05
   7  0.0152 31.2856 5.507e-05
   8 -0.0182 32.3426 8.086e-05
   9 -0.0013 32.3479 1.733e-04
  10  0.0190 33.4976 2.246e-04
```

⇒ T ⇒ sample size. ⇒ slide 52

$\hat{\gamma}(1) = -0.06$

$k=1$

$k=2$

$\hat{\gamma}(2) = -0.0013$

slide 42

very close to zero.

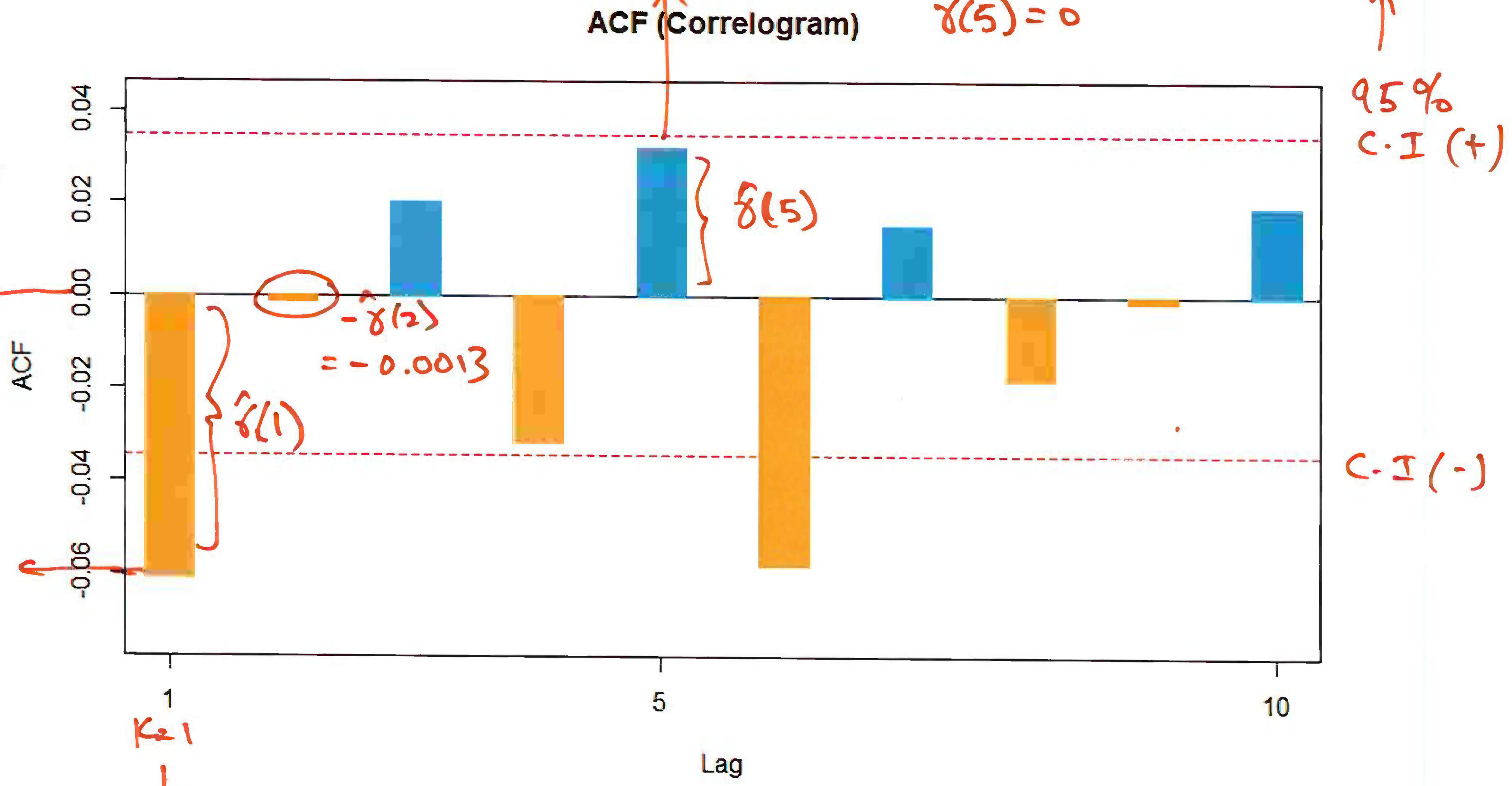
Correlogram

$\hat{\gamma}(5)$ is insignificant

$\downarrow$

$\gamma(5) = 0$

Slide 93



$\downarrow$

$\hat{\gamma}(1)$ is significant

$\hookrightarrow$ $\gamma(1) \neq 0$

Correlogram

The "Q" column represents a joint test of the significance of the autocorrelations. Specifically the k^{th} Q statistic is a test of the null hypothesis

k=2

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_k = 0,$$

$H_0 : \gamma_1 = \gamma_2 = 0$

against

$$H_1 : \text{ at least one of } \gamma_1, \ldots, \gamma_k \neq 0.$$

HA: at ....

Slide 40 ← Q = 11. 8891

p-value = 0

The last column heading "Prob" contains the p values for these Q statistics.

Reject Ho.

If the p value is less than 0.05 then we reject H_0 and conclude that there is at least one significant autocorrelation at lag k or less.

EMH fails.

Correlogram

The dashed lines on the graph represent the 95% confidence interval of the autocorrelation, which is equal to

$$\pm\frac{2}{\sqrt{T}}$$

Any autocorrelation which goes outside of the dotted lines to either side is significantly different from zero.

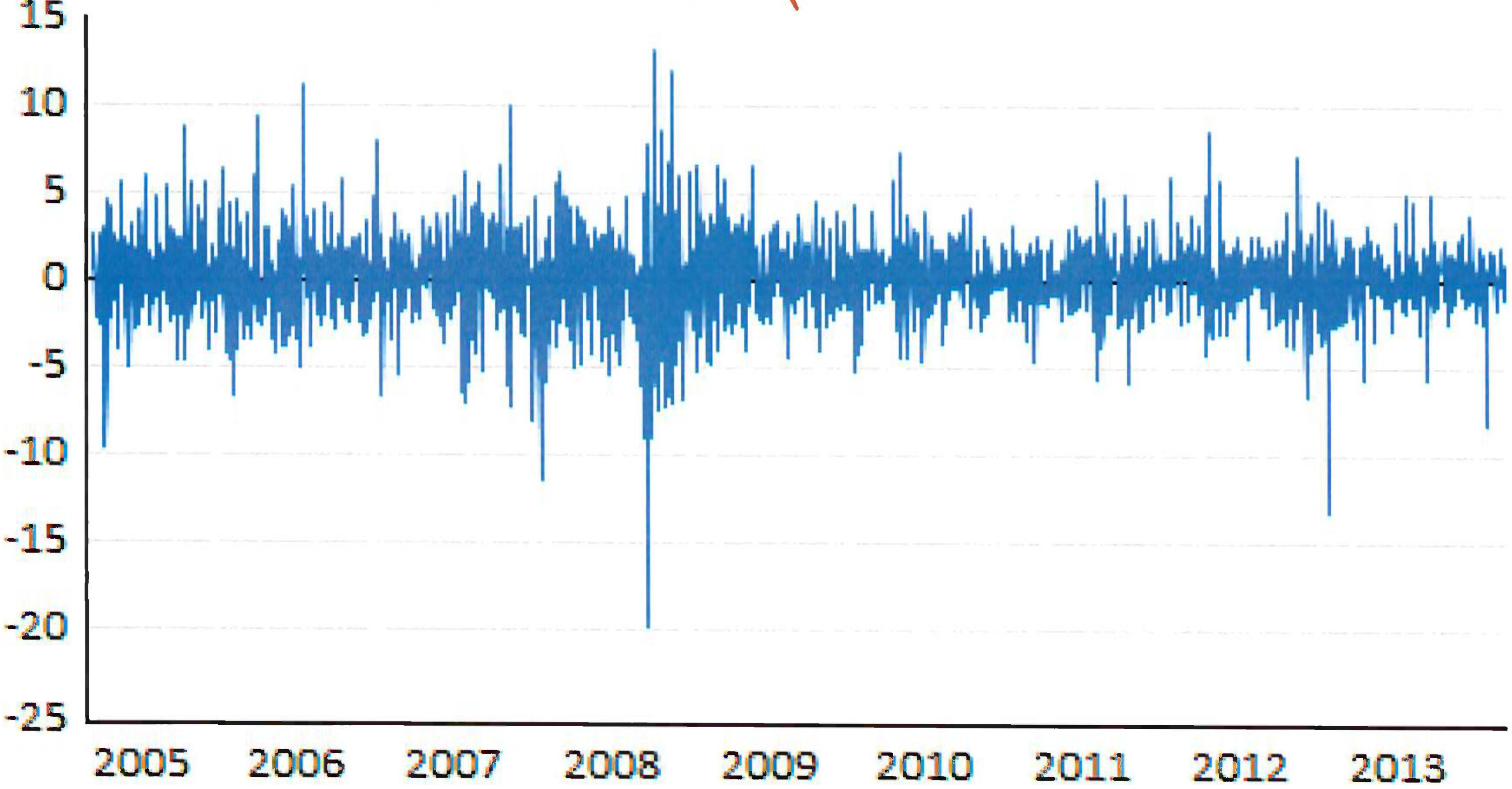
```
> # plot Apple
> plot(prices.xts)

> # log return
> lr <- na.omit(diff(log(prices.xts)))*100
```


APPLE
prices.xts
700
600
500
400
300
200
100
0
2005
2006
2007
2008
2009
2010
2011
2012
2013

APPLE_RET

lr ⇒ slide 44



Correlogram

```
> source("correlogram.R")

> length(lr)
[1] 2348
```

⇒ T ⇒ sample size. → slide 52

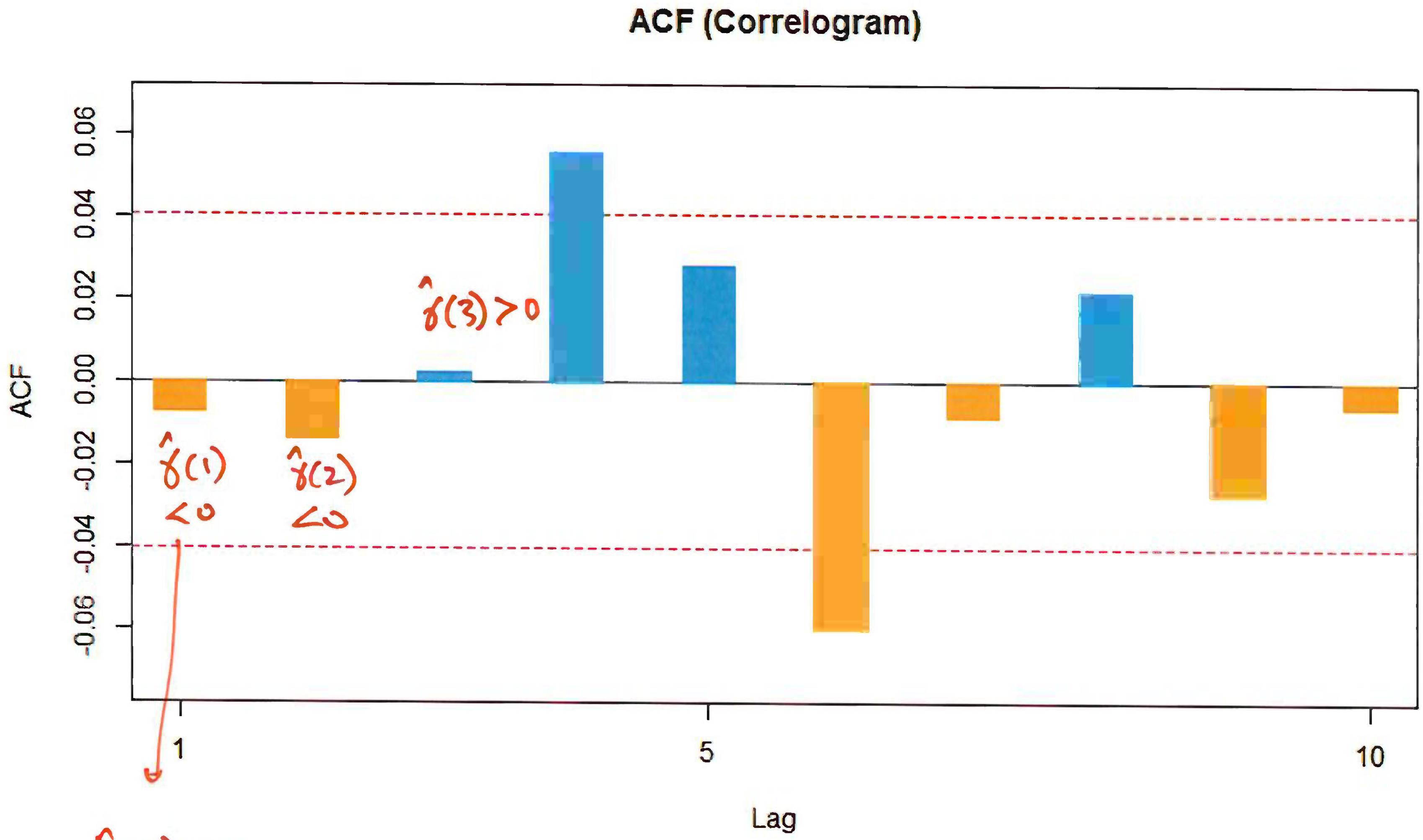
```
> correlogram(lr,max.lag = 10)
```

Lag	AC	Q	Prob
1	-0.0072	0.1230	0.725800
2	-0.0137	0.5619	0.755100
3	0.0025	0.5770	0.901700
4	0.0553	7.7662	0.100500
5	0.0284	9.6641	0.085330
6	-0.0600	18.1523	0.005863
7	-0.0085	18.3245	0.010590
8	0.0218	19.4402	0.012670
9	-0.0271	21.1726	0.011910
10	-0.0061	21.2604	0.019350

$-0.007 = \hat{\gamma}(1)$

$-0.0137 = \hat{\gamma}(2)$

even smaller than slide 40.



$\hat{\gamma}(1)$ is insignificant.

The correlogram under the column AC gives **sample autocorrelations**, of various lag lengths, for the log return of $/£ and Apple.

It is clear that all autocorrelations are close to zero. This suggests that these stocks are not predictable using their own past histories.

While the ACFs for returns look small, to be certain of the validity of the EMH, we test their significance.

Hypothesis Test:

- *If $\{y_t\}$ is stationary, when the sample size T is large, $\hat{\gamma}(k)$ should be approximately normal.*

$\sim N(0,1) \rightarrow$ c.v.

- $\hat{\gamma}(k) \sim N(0, 1/T)$, *so that* $\underbrace{\sqrt{T}\hat{\gamma}(k)}_{\text{test-statistics}}$ *will be asymptotically standard normal.*

- *Test of EMH*

$$H_0 : \gamma(k) = 0 \quad \Rightarrow \text{ EMH holds.}$$
$$H_1 : \gamma(k) \neq 0$$

- *Decision rule: reject H_0 if $|\sqrt{T}\hat{\gamma}(k)|$ is larger than 1.96, which is the critical value for the standard normal associated with a two-sided hypothesis test at the 5% significance level.*

Fama hopes fail to reject H_0.

Test of EMH

1. This hypothesis test means that a very simple statistical test of the EMH exists.

2. If $\hat{\gamma}(k)$ is statistically different from zero $\Longrightarrow$ EMH cannot be satisfied.

The EMH can be tested simply by testing the most important ACF, namely, $\hat{\gamma}(1)$. $k=1$

From the correlogram, we have

Slide 40

\$/£: $\hat{\gamma}(1) = -0.061$, Test statistic = $\underbrace{\sqrt{3189}}_{\sqrt{T}} * \underbrace{-0.061}_{\times\ \hat{\gamma}(1)} = -3.44$ C.V. $|-3.44| > 1.96$ Reject H_0 EMH fails

Slide 47

Apple: $\hat{\gamma}(1) = -0.007$, Test statistic = $\underbrace{\sqrt{2348}}_{\sqrt{T}} * \underbrace{-0.007}_{\hat{\gamma}(1)} = -0.34$ $|-0.34| < 1.96$ Fail to reject H_0

We cannot reject the EMH for Apple but we can for \$/£. This implies that \$/£ returns are predictable. EMH holds

Testing for Autocorrelation

1. We can use the LM test for autocorrelation in our analysis of the CAPM.

2. To construct such an LM test, recall the idea of an auxiliary regression model. In this case, we have the residuals being

$r_t = \mu + \varepsilon_t$ → so that ε_t has zero mean

$$\hat{\epsilon}_t = r_t - \bar{r}, \quad \underbrace{\bar{r} = \sum_{t=1}^{T} r_t/T}_{\text{sample mean}}.$$

⇒ de-mean ↓ zero mean for $\hat{\varepsilon}_t$

3. Test for autocorrelation of $\hat{\epsilon}_t$ using the auxiliary regression model

$$\hat{\epsilon}_t = \gamma_0 + \gamma_1 \hat{\epsilon}_{t-1} + \gamma_2 \hat{\epsilon}_{t-2} + \cdots + \gamma_k \hat{\epsilon}_{t-k} + v_t$$

where v_t is some random disturbance term.

$H_0: \gamma_1 = \gamma_2 = \cdots = \gamma_k = 0 \Rightarrow$ EMH holds.

- Test

$$H_0 : \gamma_1 = \cdots = \gamma_k = 0 \;\; \text{[EMH is satisfied]}$$
$$H_1 : \text{at least one differs from zero [EMH is \textbf{not} satisfied]}$$

- We can use as our test statistic $AR(k) = T \cdot R^2$ which will have a χ^2_k distribution under the null hypothesis.

- We can compare this with the corresponding χ^2_k critical value to assess whether or not we reject the null hypothesis.

Example

Let us test for serial correlation using returns on $ / £ .

- The null and alternative hypotheses are given by:

K=2

$$H_0 : \gamma_1 = \gamma_2 = 0$$

$$H_1 : \text{At least one differs from zero}$$

- From the auxiliary regression model:

after de-mean $\leftarrow$

$$\hat{\epsilon}_t = \gamma_0 + \gamma_1 \hat{\epsilon}_{t-1} + \gamma_2 \hat{\epsilon}_{t-2} + v_t.$$

- The test statistic $\mathrm{AR}(2) = T \cdot R^2 = 3189 * 0.003749 = 11.955$ > c.v.
- We will reject the null hypothesis at the 5% significance level if $\mathrm{AR}(2) > \chi^2_2(.05)$ ⇒ 5.911

  ↓ reject H₀
- We can reject the null of EMH!

```
> fit <- lm(logret ~ 1)

> summary(fit)

Call:
lm(formula = logret ~ 1)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0139146 -0.0007195 -0.0000090  0.0007195  0.0111844

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 9.05e-06   3.32e-05   0.273    0.785

Residual standard error: 0.001875 on 3188 degrees of freedom
```

$r_t = \mu + \varepsilon_t$

intercept

$\hat{\mu}$

get residuals $\hat{e}_t$

zero mean.

$\boxed{WN \quad E(e_t)=0}$

BG-LM test

```
> library(lmtest)

> bgtest(fit, order = 2, type = "Chisq")

        Breusch-Godfrey test for serial correlation of order up

data:  fit
LM test = 11.955, df = 2, p-value = 0.002535
```

< 0.05

AR(2)

slide 55

reject.

Variance ratio

$T = 10$

1-period

r_t

$t=1$ $\quad r_1$

$t=2$ $\quad r_2$

$t=3$ $\quad r_3$

$t=4$ $\quad r_4$

$\vdots$ $\quad \vdots$

$t=10$ $\quad r_{10}$

$\downarrow$

variance

$\downarrow$

s_1^2

$n = 3$-period

$$r_{3,t} = \overbrace{\underline{r_1 + r_2 + r_3}}$$

$$r_t + r_{t-1} + r_{t-2}$$

$t=1$

$t=2$

$t=3$ $\quad r_{3,3} = r_3 + r_2 + r_1$

$t=4$ $\quad r_{3,4} = r_4 + r_3 + r_2$

$\vdots$ $\quad \vdots$

$t=10$ $\quad r_{3,10} = r_{10} + r_9 + r_8$

$\downarrow$

variance

$\downarrow$

$s_n^2 = s_3^2$

$$VR_n = VR_3 = \frac{s_3^2}{n\, s_1^2}$$

An alternative way to examine the efficient markets hypothesis is to compare the variance on returns over different time horizons.

Recall the 1-period log returns

Week 1 $$r_t = \log P_t - \log P_{t-1}$$

and the n-period returns

Week 2 $$r_{nt} = \log P_t - \log P_{t-n} = r_t + r_{t-1} + \cdots + r_{t-(n-1)},$$

$n=2$

$$E(r_t + r_{t-1}) = E(r_t) + E(r_{t-1}) = \mu + \mu \doteq 2\mu$$

2-period

$$\begin{aligned} var(r_t + r_{t-1}) &= var(r_t) + var(r_{t-1}) + 2\underbrace{cov(r_t, r_{t-1})}_{=0 \text{ EMH}} \\ &= \sigma^2 + \sigma^2 \\ s_2^2 \Leftarrow &= 2\sigma^2 \end{aligned}$$

1 period $\leftarrow$ $var(r_t) = \sigma^2$

2-period

$$VR = \frac{s_n^2}{n s_1^2} = \frac{2\sigma^2}{2\sigma^2} = 1$$

1-period. EMH holds.